



APPROXIMATION FOR SUMS AND DENSITIES OF STATE OF VIBRATIONS COUPLED WITH SYMMETRIC TOP K-ROTOR

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Abstract—The energy of a symmetric top rotor depends on quantum numbers J and K . In unimolecular rate theory the energy of the K -rotor, i.e. the K -dependent part of the symmetric top rotor energy, may be considered under some conditions as “active”, i.e. mixed with vibrational energy and thus contributing to the sum and density of vibrational states of the molecule, subject to the restriction $-J \leq K \leq +J$. In these cases proper evaluation of the sum/density of vibrational plus active K -rotor states at specified energy and J involves a summation of sums/densities over all the allowed K s, a computationally intensive exercise. This work proposes simple and accurate approximations that require only trivial machine time. For the prolate symmetric top, the K -dependence of vibrational + rotational sum/density of states is approximated by a truncated Gaussian, and for the oblate top by a hyperbolic cosine. In each case the necessary parameters are obtained by a fit to smoothed “exact” vibrational sums/densities at $K = 0$ and one, or at most two, other values of K . The approximations are tested on a selection of typical symmetric top molecules at several energies and values of J with satisfactory results. The use of symmetry numbers is briefly invoked. Copyright © 1996 Elsevier Science Ltd

Key words: Sum of states, density of states, prolate/oblate symmetric top, K -rotor, vibration-rotation coupling, Gaussian approximation, RRKM, hyperbolic cosine.

1. INTRODUCTION

Many reactants in unimolecular reactions are symmetric tops in which two of the three rotational constants for overall rotation A , B , C , are equal, either exactly or at least approximately. The rotational energy E_r of such tops is characterized by quantum numbers J and K :

$$E_r(J, K) = J(J+1)B + (A-B)K^2 \quad (1)$$

where K is restricted to integral values between $-J$ and $+J$ (zero included), for a total of $(2J+1)$ values of K .

If $A > B = C$, i.e. the two *smallest* rotational constants are equal, the top is said to be prolate, the most common example of which is ethane. In a molecule of this type, Coriolis interaction causes a coupling between the angular momentum component about the top axis, as measured by the quantum number K , and the doubly degenerate vibrations (Herzberg, 1960), which in turn are coupled with all other vibrations through anharmonic interactions. This has implications for the intramolecular dynamics of the molecule, since the K -dependent part of the rotational energy in equation (1) (henceforth referred to as “ K -rotor”) can then be effectively mixed with all vibrational energy levels.

If $A < B = C$, i.e. the two *largest* rotational constants are equal, the top is said to be oblate, a common example of which is the tetrahedral CH_3 group. In such cases the K -rotor energy levels may become likewise mixed with vibrational energy levels for the same reasons as in ethane.

The RRKM Theory of Unimolecular Reactions (Robinson & Holbrook, 1972; Forst, 1973; Gilbert & Smith, 1990) is a statistical theory that expresses the microcanonical rate constant $k(E, J)$ for the decomposition of a reactant molecule having internal energy E and angular momentum quantum number J in the form

$$k(E, J) = \frac{G^*[E - E_0(J)]}{N(E, J)} \quad (2)$$

where $G^*[E - E_0(J)]$ is the sum of states for the transition state, $N(E, J)$ is the density of states for the reactant molecule and $E_0(J)$ is the energy threshold for reaction at specified J (Forst, 1973). Note that the zero of energy in equation (2) is the ground vibrational state of the reactant. RRKM is essentially a semi-classical theory so that $G^*[E - E_0(J)]$ and $N(E, J)$ are implicitly assumed to be smooth-function approximations.

The sum and density of states in equation (2) refers to so-called “active” degrees of freedom, i.e. degrees

of freedom among which energy is effectively scrambled. In this sense the K -rotor in a symmetric top with significant Coriolis interaction can be considered an "active" degree of freedom, the energy of which mixes with the energy of all vibrational degrees of freedom and thus contributes to the sum/density of states. This is what is meant by saying that K may not be a "good" quantum number, in the sense that it is not a constant of motion that conserves the same value throughout the reaction (Aubanel *et al.*, 1991).

In cases where K is not a "good" quantum number, it is usually assumed for simplicity that the K -rotor is "separable". By "separable" is meant that the J - and K -dependent parts of $E_r(J, K)$ can be dealt with separately (Current & Rabinovitch, 1963; Forst & St. Laurent, 1967; Miller, 1979; Smith, 1993). As a further simplification in the RRKM treatment of large molecules, it has been the custom to ignore the restriction on the quantum number K in the active K -rotor, which means that the corresponding rotational energy

$$E_r(K) = (A - B)K^2 \quad (3)$$

is assumed to be limited only by total available energy rather than by the specified J (Forst, 1973; Gilbert & Smith, 1990; Forst, 1991; Zhu *et al.*, 1993). This greatly simplifies the calculation of the sum/density of vibrational-rotational states and produces no serious error for very high values of J , but causes an overcount at lower values of J , which can of course be corrected in principle by executing a separate count for each value of K between $-J$ and $+J$. However this increases considerably the computational requirements in a typical rate calculation, so that a reasonably fast and accurate method would be helpful.

This work is addressed specifically to the problem of finding a reasonably accurate approximation for the determination, at specified J , of the density/sum of vibrational states coupled with active K -rotor rotational states that respect the restriction $-J \leq K \leq +J$.

2. THEORY

The total *active* energy E of a symmetric top molecule at specified J and K with an active and separable K -rotor is the sum of its internal vibrational energy (E_v) and rotational energy of the K -rotor [$E_r(K)$]:

$$E = E_v + E_r(K). \quad (4)$$

Note that E defined here is not the *total*, i.e. active + "inactive" energy E_{tot} of the molecule, the "inactive" part being the energy of the J -rotor $E_r(J) = J(J+1)B$; thus $E_{\text{tot}} = E_v + E_r(J, K)$. In what follows E is considered to be a constant having specified value, so that the development is equally valid if this constant is interpreted as E_{tot} .

The "exact" sum/density of states at specified E, J within the active K -rotor approximation is obtained by calculating the sum of a sequence of vibrational sum/density of states at $E_v = E - E_r(K)$, for $K = 0, \pm 1, \pm 2, \dots, K_{\text{max}}$, where K_{max} is the nearest integer determined by the requirement that $E - E_r(K)$ remains positive. Thus the sum of vibrational + K -rotor states $G_{vr}(E, J)$ at specified E, J is

$$G_{vr}(E, J) = \sum_{K=-K_{\text{max}}}^{K_{\text{max}}} G_v[E - E_r(K)] \quad (5)$$

that on account of the symmetry in the K -values can be written

$$G_{vr}(E, J) = G_v(E) + 2 \sum_{K=1}^{K_{\text{max}}} G_v[E - E_r(K)] \quad (6)$$

and similarly the corresponding density of states is

$$N_{vr}(E, J) = N_v(E) + 2 \sum_{K=1}^{K_{\text{max}}} N_v[E - E_r(K)]. \quad (7)$$

The present problem is to avoid the summations over individual K s in equations (6) and (7). Since these summations have different characteristics in the case of prolate or oblate tops, they require different approximations, which will be discussed separately.

2.1. Prolate top

In this case $A - B$ is positive, so that the condition for $E - E_r(K) \geq 0$ requires that K does not exceed K_{max} defined as the smaller of J and $+\sqrt{E/(A - B)}$:

$$K_{\text{max}} = \min\{J, +\sqrt{E/(A - B)}\}. \quad (8)$$

With this restriction, if the individual terms of the sum in equation (5) or in the analogous sum for the density $N_{vr}(E, J)$ are plotted up as a function of K , using a smooth-function approximation for $G_v[E - (A - B)K^2]$ or $N_v[E - (A - B)K^2]$, the result for both is very nearly a discrete truncated Gaussian centred at the origin. This is shown in Figs 1 and 2 as a smooth curve, obtained by joining terms with adjacent values of K by a straight line.

Such a Gaussian centred at the origin can be written

$$g(K) = g(0) e^{-\alpha K^2}. \quad (9)$$

In the present case the variable K can take up only the discrete values $0, \pm 1, \pm 2, \dots, \pm K_{\text{max}}$. Suppose n is one of the discrete K -values, thus

$$-\alpha = \frac{1}{n^2} \ln \frac{g(n)}{g(0)} \quad (10)$$

and therefore for any K

$$g(K) = g(0) \exp \left\{ \frac{1}{n^2} \ln \frac{g(n)}{g(0)} \right\} K^2 \quad (11)$$

which means that the entire Gaussian can be reconstructed from just two points, the maximum $g(0)$ at $K = 0$ and another value $g(n)$ at $K = n$.

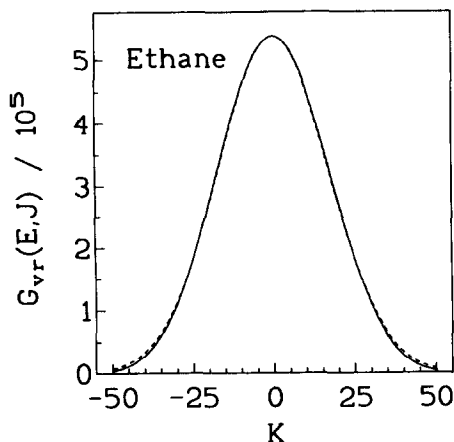


Fig. 1. Smooth-function representation of individual terms in the series for the sum of vibrational + K -rotor states $G_{vr}(E, J)$ in prolate ethane at $E = 10,000 \text{ cm}^{-1}$, $J = 50$. Continuous line: exact count, equation (6); dashed line: Gaussian approximation of equation (12) using $\alpha = -0.0017375$.

Applied to the case of a prolate top, we have

$$\frac{G_v(E, J)}{N_v(E, J)} = \frac{G_v(E)}{N_v(E)} \times \left\{ 1 + 2 \sum_{K=1}^{K_{\max}} \exp(\alpha K^2) \right\} \quad (12)$$

where

$$-\alpha = \frac{1}{n^2} \ln \frac{g[E - (A - B)n^2]}{g(E)} \quad (13)$$

and g stands for $G_v[\dots]$ or $N_v[\dots]$. The individual terms in the summation of equation (12) for two specific cases are shown (from $-K_{\max}$ to $+K_{\max}$) as dashed lines in Figs 1 and 2.

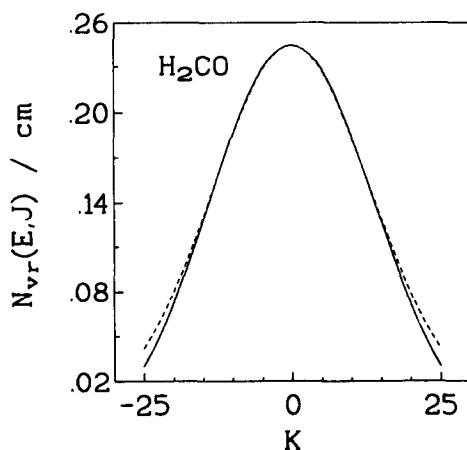


Fig. 2. Smooth-function representation of individual terms in the series for the density of vibrational + K -rotor states $N_{vr}(E, J)$ [cm^{-1}] $^{-1}$ in prolate formaldehyde at $E = 10,000 \text{ cm}^{-1}$, $J = 25$. Continuous line: exact count, equation (7); dashed line: Gaussian approximation of equation (12) using $\alpha = -0.0028719$.

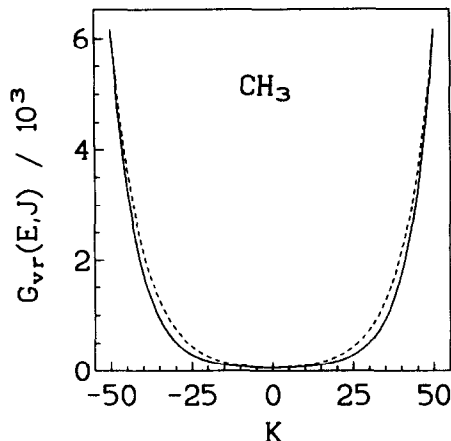


Fig. 3. Smooth-function representation of individual terms in the series for the sum of vibrational + K -rotor states $G_{vr}(E, J)$ in oblate CH_3 at $E = 5000 \text{ cm}^{-1}$, $J = 50$. Continuous line: exact count, equation (6); dashed line: one-parameter hyperbolic cosine approximation of equation (20) using $a = 0.10531$, $b = 1$. The result of the two-parameter approximation with $a = 0.047273$, $b = 1.2048$ is indistinguishable from the exact count.

2.2. Oblate top

Contrary to the prolate top, here $A - B$ is negative; thus, in the oblate top $E - E_r(K)$ is positive, so that K is not constrained by the value of E . However, since each K reduces the energy of the symmetric top [equation (1)], $K_{\max}$ is constrained in principle by the requirement that $E_r(J, K)$ in equation (1) remains positive, i.e. that $K_{\max}$ be the integer nearest

$$K_{\max} = \pm \sqrt{\frac{J(J+1)B}{B-A}}. \quad (14)$$

If we write the term under the radical in the form $J(J+1)/(1 - A/B)$, we see that equation (14) amounts to $K_{\max} \cong \pm J$ since in the oblate top A/B is always less than unity, so that there is effectively no constraint other than the value of J .

Figure 3 shows a plot of the individual terms in the summation of equation (5) for the sum of states of the CH_3 oblate top, and Fig. 4 shows individual terms in a similar summation for the density of states of the BF_3 oblate top, represented again in each case as a smooth curve, obtained by joining terms with adjacent values of K by a straight line.

The form of the curves is somewhat different (owing to numerically different rotational constants), but in fact both can be approximated by a hyperbolic cosine function of the form $\cosh(aK^b)$ where a and b are constants. Since $\cosh(0) = 1$, let us first define for this purpose

$$h(K) = \frac{H[E - E_r(K)]}{H(E)} \quad (15)$$

where H represents $G_v[\dots]$ or $N_v[\dots]$; then $h(K=0) = 1$. Given that the largest contributions to the sums in equations (6) and (7) come from large K

(cf. Figs 3 and 4), we may first try a simple fit at $K_{\max}$ with $b = 1$ by letting $h(K_{\max}) = \cosh(aK_{\max})$ from which

$$a = \frac{\cosh^{-1}[h(K_{\max})]}{K_{\max}} \quad (16)$$

Tests have shown that in some cases an accurate representation of $h(K)$ requires two parameters, a and b , to be evaluated by a fit to exact values of $H[E - E_r(K)]$ at two different K s, one of which may be $K_{\max}$ as above, and some other K_i . Thus

$$\begin{aligned} h(K_{\max}) &= \cosh(aK_{\max}^b) \\ h(K_i) &= \cosh(aK_i^b) \end{aligned} \quad (17)$$

from which by inversion

$$b = \frac{\ln \left(\frac{\cosh^{-1}[h(K_{\max})]}{\cosh^{-1}[h(K_i)]} \right)}{\ln \left(\frac{K_{\max}}{K_i} \right)} \quad (18)$$

and

$$a = \frac{\cosh^{-1}[h(K_{\max})]}{K_{\max}^b} = \frac{\cosh^{-1}[h(K_i)]}{K_i^b} \quad (19)$$

The approximation to the sum or density of states of an oblate top is thus

$$\frac{G_{vr}(E, J)}{N_{vr}(E, J)} = \frac{G_v(E)}{N_v(E)} \times \left\{ 1 + 2 \sum_{K=1}^{K_{\max}} \cosh(aK^b) \right\} \quad (20)$$

which requires at most three accurate determinations of $G_v[E - E_r(K)]$ or $N_v[E - E_r(K)]$, at $K = 0$, $K_{\max}$ and one other K . The individual terms in the summation

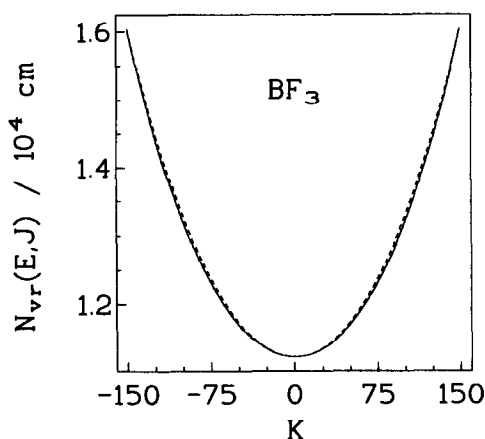


Fig. 4. Smooth-function representation of individual terms in the series for the density of vibrational + K -rotor states $N_{vr}(E, J)$ in oblate BF_3 at $E = 50,000 \text{ cm}^{-1}$, $J = 150$. Continuous line: exact count, equation (7); dashed line: one-parameter hyperbolic cosine approximation of equation (20) using $a = 0.005976$, $b = 1$. The result of the two-parameter approximation with $a = 0.0040884$, $b = 1.0756$ is indistinguishable from the exact count.

of equation (20) are shown (from $-K_{\max}$ to $+K_{\max}$) as dashed lines in Figs 3 and 4 for two specific cases.

3. RESULTS

The general procedure consisted of calculations at $E = 5000, 10,000, 20,000$ and $50,000 \text{ cm}^{-1}$ and $J = 10, 20, 50$ and 100 at each energy. This choice of E, J combinations was deemed sufficiently representative to evaluate the performance of the approximations. For each molecule, and at each E, J , both the sum and density of states were calculated "exactly" and by the approximation formula, equation (12) or equation (20) as the case may be, and compared. The "exact" calculation was a smooth-function version of the exact count generated by the method of steepest-descent (Forst, 1973) which has been shown to be an excellent representation. Error in % is defined as $100 \times (1 - \text{"exact"/approximate})$. Molecular parameters of the molecules examined are listed in the Appendix.

3.1. Prolate tops

The following cases were examined: ozone, formaldehyde, CF_3O_2 and ethane. These represent molecules of increasing complexity, the last being a symmetric top exactly and the others approximately so. In these cases an average rotational constant B was used, defined as the square root of the product of the two nearly equal rotational constants [this is a geometric average; some authors prefer an arithmetic average (Zhu & Hase, 1990), which is only slightly different].

If the K -dependence of the sum or density of states as given by equation (12) were a perfect Gaussian, α would be independent of the choice of n . This is not quite the case, so that there is some drift of α with n as shown in Table 1. The data suggest that in general best results are obtained by choosing n as the integer nearest $\frac{1}{2}K_{\max}$, which is the case of the dashed lines in Figs 1 and 2.

For the two largest molecules, CF_3O_2 and ethane, comparison shows equation (12) to yield results for both the sum and density of states, at all energies and all J examined, to within better than 1%. The worst result for CF_3O_2 is the sum of states at $E = 5000 \text{ cm}^{-1}$, $J = 100$ with a deviation of 0.4%, and for ethane the density at $J = 50$ and $E = 10,000$ and $20,000 \text{ cm}^{-1}$, with a deviation of 0.6%. For formaldehyde, the approximation of equation (12) deviates likewise mostly by about 1%, the worst results being the sum and density of states at $J = 50$ and 100 with a deviation of about 2%. The same is true of ozone, except that at $J = 50$ and 100 some of the sum and density of states deviate by about 5%. There is thus an obvious trend towards better results as the molecule examined becomes larger.

In general, for about $J \geq 50$, the contribution of terms with $K \geq 50$ is quite small, so that the restriction $-J \leq K \leq +J$ has little effect and the result for the sum or density of states is very close to

Table 1. Values of α in equation (13) for various n in several prolate tops

E, cm^{-1}	J	n	$-\alpha \times 10^3$	% error*
ethane $G_{\text{vr}}(E, J)$				
10,000	50	1	1.6751	2.2
		12	1.6889	1.9
		16	1.6997	1.6
		25	1.7375	0.5
		33	1.7891	-0.9
ethane $N_{\text{vr}}(E, J)$				
20,000	75	1	1.0302	2.3
		18	1.0413	1.8
		25	1.0519	1.3
		37	1.0797	-0.02
		50	1.1265	-2.2
CF_3O_2 $N_{\text{vr}}(E, J)$				
1000	25	1	0.22044	0.025
		6	0.22055	0.023
		8	0.22064	0.021
		12	0.21089	0.016
		16	0.22125	0.009
Formaldehyde $N_{\text{vr}}(E, J)$				
10,000	25	1	2.703	3.4
		6	2.7297	3.1
		8	2.7516	2.8
		12	2.8169	1.9
		16	2.9159	0.65
Ozone $G_{\text{vr}}(E, J)$				
50,000	100	1	0.18288	6.2
		25	0.18644	5.5
		33	0.18922	5.0
		50	0.19840	3.2
		66	0.21254	0.6

*Negative error means undercount with respect to exact result.

what would be obtained by assuming that $E_r(K)$ is limited only by the available energy, in which case $G_{\text{vr}}(E, J)$ or $N_{\text{vr}}(E, J)$ can be obtained directly from the steepest-descents routine (Forst, 1973), which obviates convoluting vibrations with rotations.

Some of the results shown in Table 1 suggest that in certain cases still better results could be obtained with n larger than $\frac{1}{2}K_{\text{max}}$, although this is not universally true (cf. ethane density at $E = 20,000 \text{ cm}^{-1}$, $J = 75$).

3.2. Oblate tops

The two tops examined were CH_3 and BF_3 ; they are both of comparable complexity but with numerically different rotational constants which cause a somewhat different K -dependence of the sum/density of states (cf. Figs 3 and 4).

For BF_3 , the two-parameter approximation of equation (20) produces results with a deviation generally below 0.1%, the worst result being the sum of states at $E = 5000 \text{ cm}^{-1}$, $J = 100$ with a deviation of 0.5%. For this molecule, even the one-parameter approximation yields reasonably good results, generally better than 1%, the worst case being the density and sum of states at $E = 5000 \text{ cm}^{-1}$, $J = 100$, with a deviation of 3% for the density and 4% for the sum.

Numerical values of the constants in equation (20) are shown in Table 2 for CH_3 . Here the results are similar but the deviations are larger. For the two-parameter approximations, the largest deviations occur for the density at $J = 100$ and $E = 5000$ and $10,000 \text{ cm}^{-1}$: 8 and 6%, respectively. This is a peculiar case because at these E, J the one-parameter approximation actually gives a better result: 5 and 3% deviation, respectively (Table 2). Similar results are obtained for the sum of states at the same E, J . However at $E = 50,000 \text{ cm}^{-1}$ and $J = 100$ the one-parameter approximation deviates by 13% for the density and 17% for the sum of states, while the two-parameter approximation is in error by less than 1% (Table 2).

Overall, then, equation (20) with three reference K s determines the value of $G_{\text{vr}}(E, J)$ or $N_{\text{vr}}(E, J)$ to within better than 1% in most cases. For lesser accuracy, the single parameter approximation [equation (16)] is often quite sufficient; in fact, as shown in Table 2, the parameter b in the two-parameter approximation is then close to unity.

Table 2. Value of constants in equation (20) for CH_3 oblate top*

$G_{\text{vr}}(E, J)$						
E, cm^{-1}	J	$a_1 \times 10^2$	error (%)	$a_2 \times 10^2$	b	error (%)
5000	10	7.7726	-0.2	7.2588	1.0297	-0.05
	20	8.438	-2.9	6.0623	1.1104	-0.46
	50	10.531	-16.0	3.9664	1.2496	1.1
	100	10.981	5.2	8.4252	1.0563	9.5
50,000	10	3.2206	-0.01	3.1776	1.0058	0.002
	20	3.2808	-0.13	3.0468	1.0247	-0.02
	50	3.6552	-3.9	2.1730	1.1329	-0.48
	100	4.3926	-17.3	1.3044	1.2637	0.18
$N_{\text{vr}}(E, J)/(\text{cm}^{-1})^{-1}$						
E, cm^{-1}	J	$a_1 \times 10^2$	error (%)	$a_2 \times 10^2$	b	error (%)
5000	10	6.9299	-0.14	6.5776	1.0227	-0.03
	20	7.3859	-1.8	5.7161	1.0855	-0.29
	50	8.8944	-12.9	3.8876	1.2116	0.58
	100	9.1457	5.0	7.4644	1.0441	8.3
50,000	10	2.9344	-0.01	2.9038	1.0045	-0.001
	20	2.977	-0.08	2.8105	1.0192	-0.01
	50	3.2413	-2.5	2.1515	1.1048	-0.31
	100	3.7739	-13.0	1.3824	1.2181	-0.25

* $a_1 = a$ in one-parameter approximation ($b = 1$); $a_2 = a$ in two-parameter approximation [equations (18) and (19)]. Negative error means undercount with respect to exact result.

4. DISCUSSION

The sort of accuracy attainable with equations (12) and (20) obtained above seems more than sufficient for the intended purpose, considering that the development neglects that most molecules are not (exactly) symmetric tops, and that the efficiency of Coriolis coupling responsible for the destruction of K as a "good" quantum number seems to depend on both total energy and J (Dai *et al.*, 1985; Polik *et al.*, 1990; McCoy *et al.*, 1991) even in cases where the coupling does operate. Finally the assumption of separability of the K -rotor is in itself a considerable approximation.

With this proviso, the approximations represented by equations (12) and (20) shows a respectable performance, given their simplicity. There is of course no guarantee that there may exist an exceptional pathological case that these approximations are unable to handle. Olzmann & Troe (1994) (see also Troe, 1983) propose somewhat more complicated approximations based on the Whitten–Rabinovitch approximation (Whitten & Rabinovitch, 1963) for the density of states. Their results are similar but not directly comparable since they define E as E_{tot} and calculate the density at $E_{\text{tot}} - E_r(J)$.

The approximations proposed here require two or three accurate determinations of the sum or density of states at two or three values of K . However in a rate calculation one determination is required in any event even if the $-J \leq K \leq +J$ constraint is ignored, so that the additional calculations imposed by the proposed approximations are minimal. With respect to an exact separate count at each allowed K , these approximations offer little computational advantage at low J , say, $J \leq 5$ (and therefore $K \leq 5$), but at larger J machine time saving is considerable since the sort of summation to be executed in equations (12) and (20) is a trivial matter (note that because of the symmetry in K -values, it is sufficient to sum from $K = 1$ to K_{max} rather than from $-K_{\text{max}}$ to $+K_{\text{max}}$). If the inverse hyperbolic cosine is not implemented in the software as a primitive, it can be replaced by its logarithmic equivalent, whereby $\cosh^{-1}(x) = \ln[x + (x^2 - 1)^{1/2}]$.

The development given here assumes implicitly that vibrations are harmonic. Note that with the steepest-descents routine used in the examples, anharmonic sum or density of states for independent Morse oscillators can be easily generated and substituted in equations (12) and (20) if the requisite data are available, with little additional complication and good accuracy (Forst, 1994).

RRKM theory postulates the intervention of a transition state, difficult to define precisely except by elaborate calculations. However, if the molecule fragments by simple bond fission, it is reasonable to expect that the transition state will be "loose", i.e. highly prolate: for example, a molecule of the type $X_3C \cdot CY_3$, upon sufficient extension the C–C bond will

become an almost exactly prolate symmetric top: $X_3C \cdot CY_3 \rightarrow X_3C \cdots CY_3$ even if the reactant itself is not. It can then be assumed that conditions are favourable for the K -rotor in the transition state to become active, which would justify the state count to be done by the approximation proposed here. However, in most cases, the fate of the K -rotor energy is not entirely clear and in principle several possibilities may be envisaged for the microcanonical rate constant according to whether the K -rotor is assumed to be "active" in the reactant molecule, or transition state, or both (Zhu & Hase, 1990; Zhu *et al.*, 1993).

Nothing has been said so far about symmetry considerations in connection with rotations. If the molecule in question possesses an element of symmetry, not all values of the quantum number K are necessarily present. For example, if the nuclear spin of identical atoms is zero and the molecule belongs to the symmetry group C_{2v} (e.g. formaldehyde and ozone), only one half of all K values appears in the rotational partition function (Herzberg, 1960, p. 507). The usual procedure is to execute calculations with all J -allowed values of K and then divide the result by a symmetry number σ (Quack & Troe, 1974). Thus $\sigma = 2$ in the example above; for another example take ethane with frozen internal rotation (symmetry D_{3h}) for which $\sigma = 6$ ($\sigma = 18$ if internal rotation is free).

Now if E in the equations above is interpreted as total energy E_{tot} , summing the corresponding density of states over all E, J, K must correctly reproduce the vibrational–rotational partition function Q_{vr} of the symmetric top molecule:

$$Q_{\text{vr}} = \frac{1}{\sigma} \int_0^\infty dE \left\{ \sum_{J=0}^\infty (2J+1) \times \sum_{K=-K_{\text{max}}}^{K_{\text{max}}} N_v[E - E_r(J, K)] \right\} e^{-E/kT} \quad (21)$$

where k is Boltzmann's constant and T is temperature. The point of displaying equation (21) is that it contains perforce the symmetry number σ . Note that it is immaterial whether the division by σ is affected before or after integration with respect to E . Thus in an RRKM calculation, σ has to be accounted for whether it be microcanonical (summation over J, K at fixed E) or canonical (summation over E, J, K); this goes for the density as well as for the sum of states (replacing $N_v[\dots]$ by $G_v[\dots]$ in equation (21) merely yields Q_{vr} multiplied by kT) (Forst, 1973).

For convenience, in the case of separable K -rotor, the symmetry number may be thought of as the product of two parts, the "symmetry number" σ_J of the J -rotor, and the "symmetry number" σ_K of the K -rotor, such that $\sigma = \sigma_J \sigma_K$. For example, in eclipsed ethane with $\sigma = 6$, $\sigma_J = 2$, $\sigma_K = 3$, the latter justified by noting that the K -rotor involves mainly the

rotational constant $A(A > B)$, which refers to rotation about the C-C bond which has threefold symmetry. Similar separation can be applied in the course of an RRKM calculation (Forst, 1991).

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APPENDIX

Molecular parameters (all in cm^{-1})

FR = frequencies (degeneracies in parentheses); A , B = rotational constants.

- C_2H_6^a : FR = 2954 2896 2985(2) 2969(2) 1388 1379 1472(2) 1469(2) 1206(2) 995 821(2) 289; $A = 2.671$, $B = 0.6631$
- CF_3O_2^b : FR = 1292 1270 1129 728 596 584 1100 398 380 469 870 70; $A = 0.182$, $B = 0.104$ [$= (0.104 \times 0.1045)^{1/2}$]
- H_2CO^c : FR = 2766 1746 1501 2843 1247 1164; $A = 9.412$, $B = 1.212$ [$= (1.3 \times 1.13)^{1/2}$]
- O_3^c : FR = 1110 705 1043; $A = 3.554$, $B = 0.419$ [$= (0.4453 \times 0.39477)^{1/2}$]
- CH_3^d : FR = 3044 3162(2) 1396(2) 580; $A = 4.79$, $B = 9.581$
- BF_3^e : FR = 888 697 1463(2) 481(2); $A = 0.1732$, $B = 0.3464$.

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